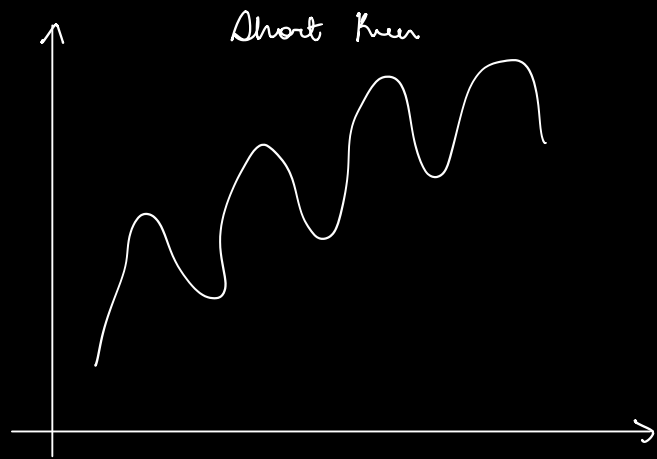
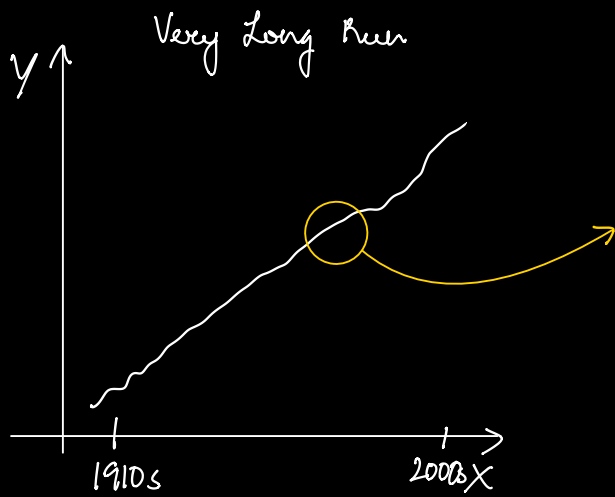


HS125



Booms and recessions are short term concepts.

Growth refers to growth of output (goods and services)
economy is doing well
GDP/GNP/LNP

Difference between a growth rate of 2% and 4% accumulates over a long period of time and lead to significantly different lifestyles.

Eg:- Japan (Growth rate of Japan after WWII - 8%)
↓ Contrast
Ghana (0% growth rate)

Reason behind

this huge difference between nations over the same period of time

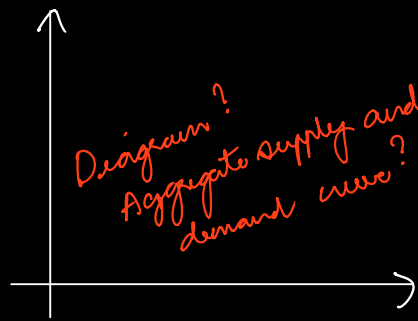
DEVELOPING NATION

DEVELOPED/UNDERDEVELOPED
NATION

① Priorities:- Wealth accumulation,
tech...

Infrastructure

Why Inflation?



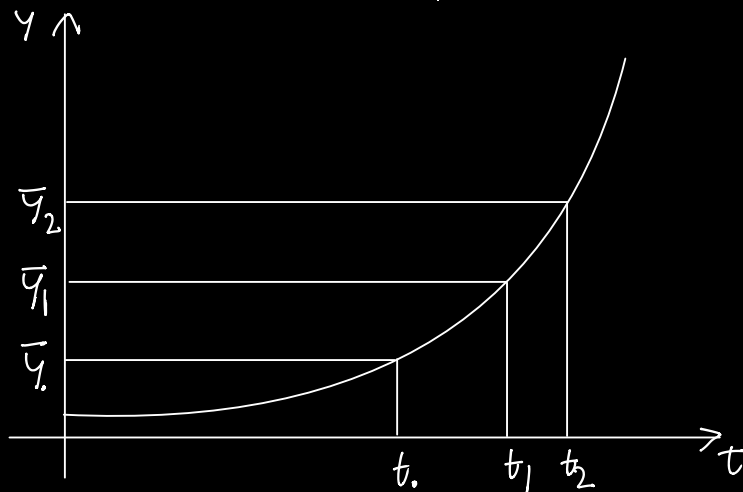
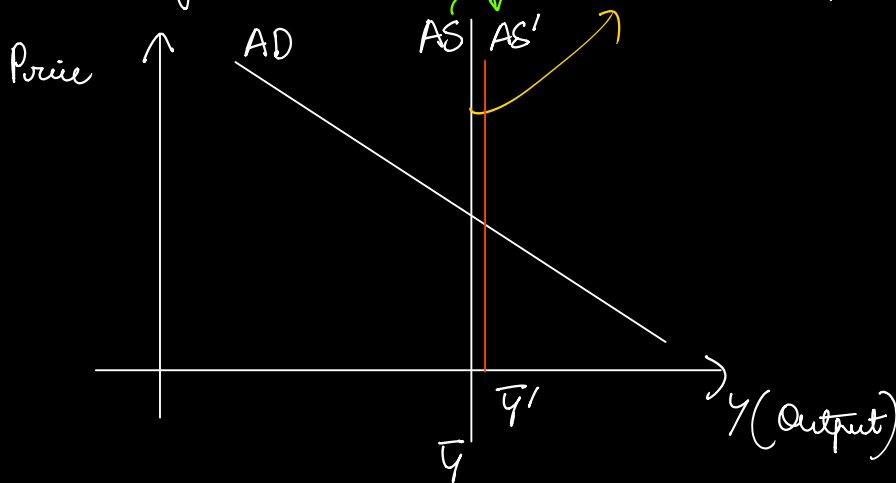
Aggregate supply curve

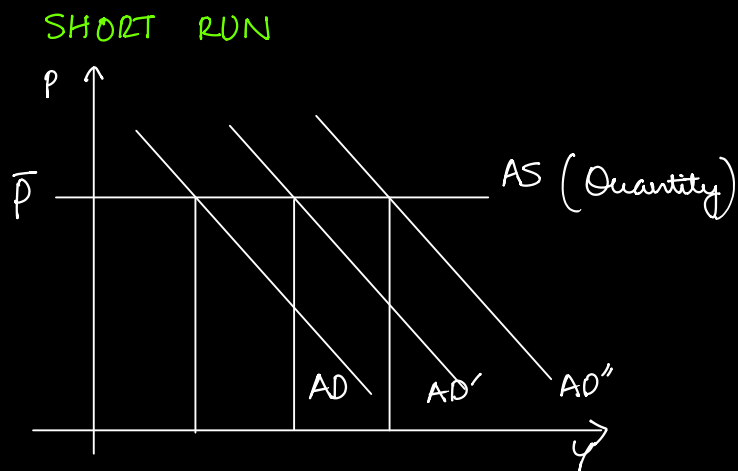
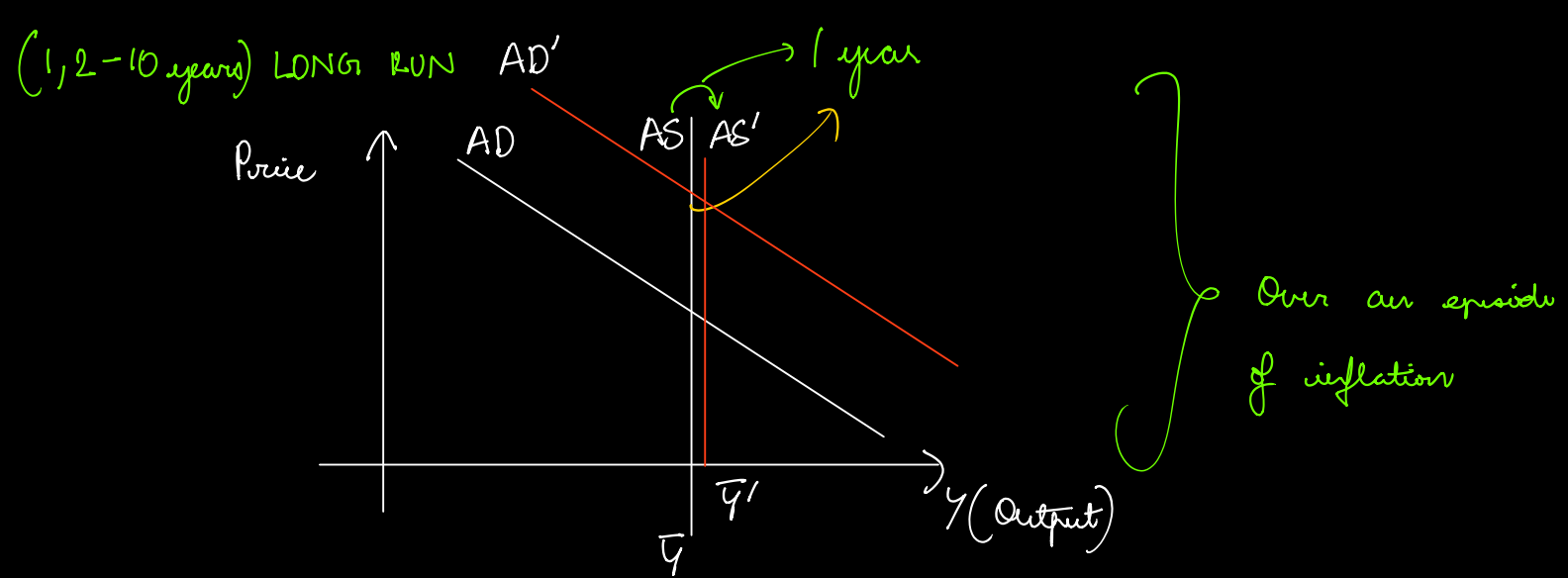
The amount of goods and services companies would produce at any price level

Aggregate demand curve

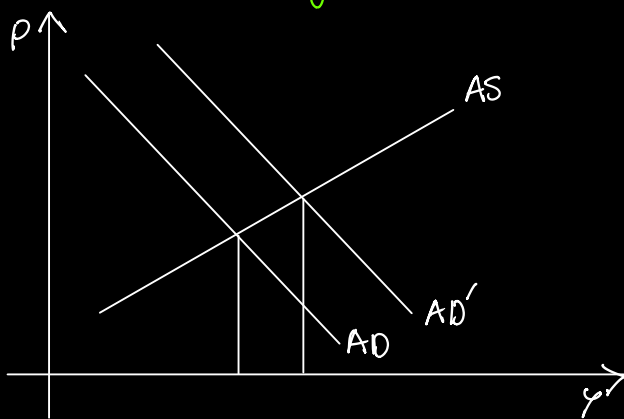
The amount of goods and services consumers would consume at any price level

Equilibrium :- Clearing of markets
 Supply = demand



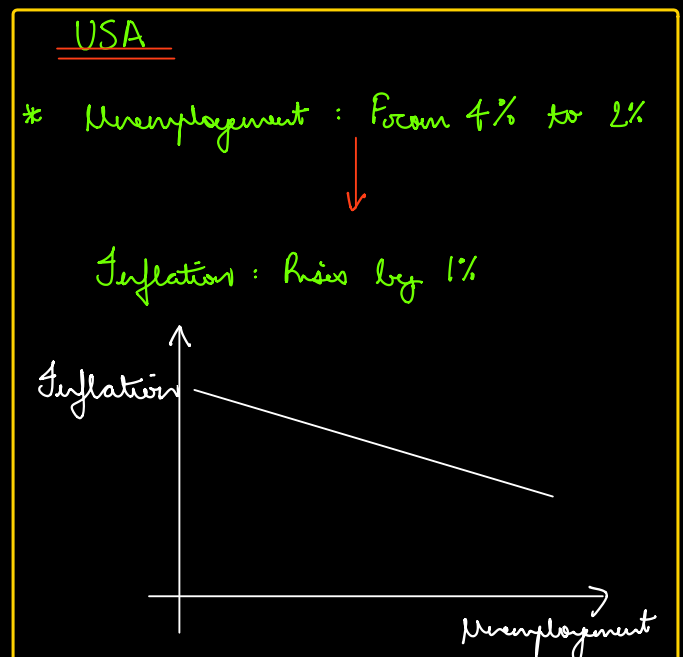


MEDIUM RUN (1 year)



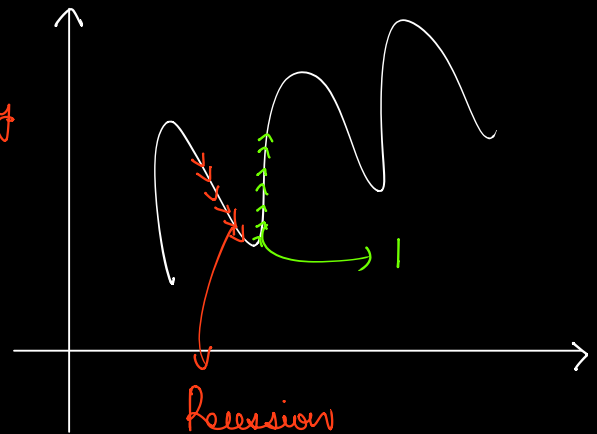
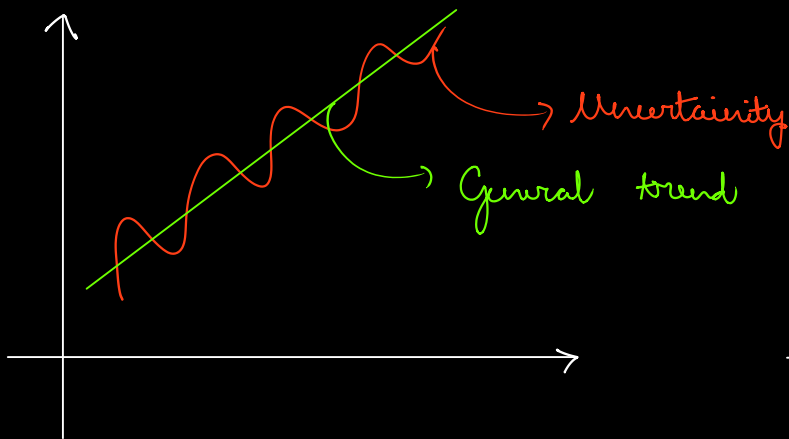
* Prices are slow to adjust

* In short to medium run AD drives the economy



Economic Growth: Growth in output
 Real output → GDP

	1913	1980	2000s
Ghana	Income	Income $\times 1.2$	Growth rate of both countries are almost the same
Brazil	Income	5x Income	the same



Output Gap = Actual output - Potential output

Boom → Output Gap ↑↑

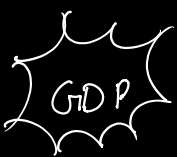
Recession → " ↓↓

22/1/25

* Mankiw

* Dornbusch, Fischer, Sturza

} → BOOKS



$$Y = F(\text{Labour}, K(\text{Capital}))$$

Wage rate (w)

Rental rate (r)

$$\text{Amount spent (Cost)} = w \times L + r \times K$$

Income that output generates

$$Y = wL + rK + \text{Profit}$$

DEFINITIONS

* GDP → Ownership is ignored, geographic location of production is more important.

* GNP: Production done in China will contribute to the GNP of US unlike GDP.

Eg: - NIKE, a US based company does a lot of its manufacturing in China.

$$GNP = GDP + \text{net income from abroad} - \text{income contributed to other countries}$$

* NDP: Net domestic product = GDP - capital depreciation

Indirect Taxes $\approx 0.1 \times NDP$

* NI = NDP - Indirect Taxes

Depreciation $\approx 0.11 \times GDP$

→ National Income
Factor payments take place from NI

$Y = AD$ (Equilibrium) → Ideal and rare

$$AD = \text{Consumption} + \text{Investment} + \text{Government Expenditure} + (X - M)$$

→ Down for increased employment

To cater to demand we must produce.

← Net exports
Demand of our goods by foreigners

Spending $\uparrow\uparrow$ (By rest of society) $\rightarrow$ Income $\uparrow\uparrow$ (Business)

Can afford to cut back on production

$$\text{Unplanned Inventory (IU)} = Y - AD$$

$IU > 0 \Rightarrow$ Inventory $\uparrow$
Investment

$IU < 0 \Rightarrow$ Inventories would be run down

bare minimum
consumption (to survive)

$$C = \bar{C} + cY$$

marginal propensity to consume

Marginal Propensity to Consume: If income is increased by 1 unit what fraction of it goes into consumption.

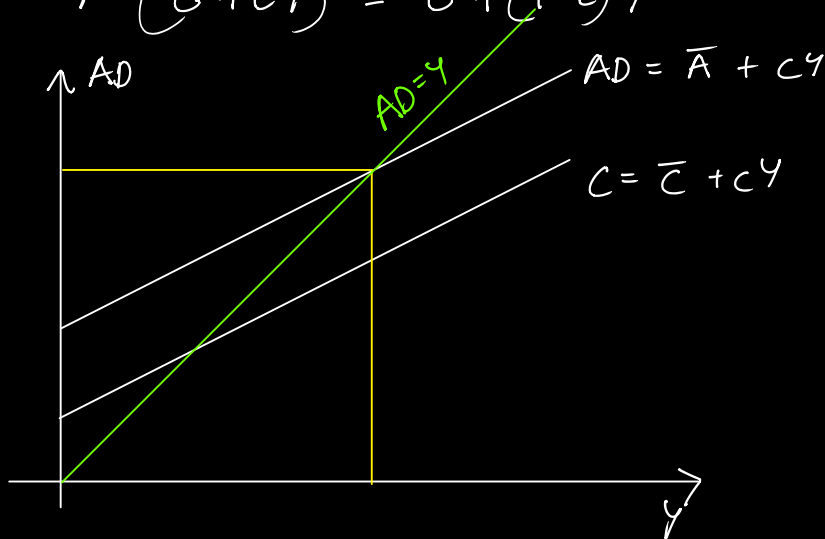


C : Consumption

$1 - C$: Savings

$$S = Y - C$$

$$= Y - (\bar{C} + cY) = \bar{C} + (1 - c)Y$$



$$\begin{aligned} AD &= C + (\bar{I} + \bar{G} + \bar{N}X) \\ &= \bar{C} + cY + (\bar{I} + \bar{G} + \bar{N}X) \\ &= cY + \underbrace{(\bar{C} + \bar{I} + \bar{G} + \bar{N}X)}_{\bar{A}} \end{aligned}$$

23/1/25

Consumption AD & Autonomous Spending

Components of AD

1. Investment = $\bar{I}$
2. Govt Spending = $\bar{G}$
3. Taxes = $\bar{T}_A$
4. Transfers (subsidies) = $\bar{T}_R$
5. Net Exports = $\bar{N}_X$

form of social or unemployment benefits

Why are exports imp?

Greater demand of our goods among foreign consumers.

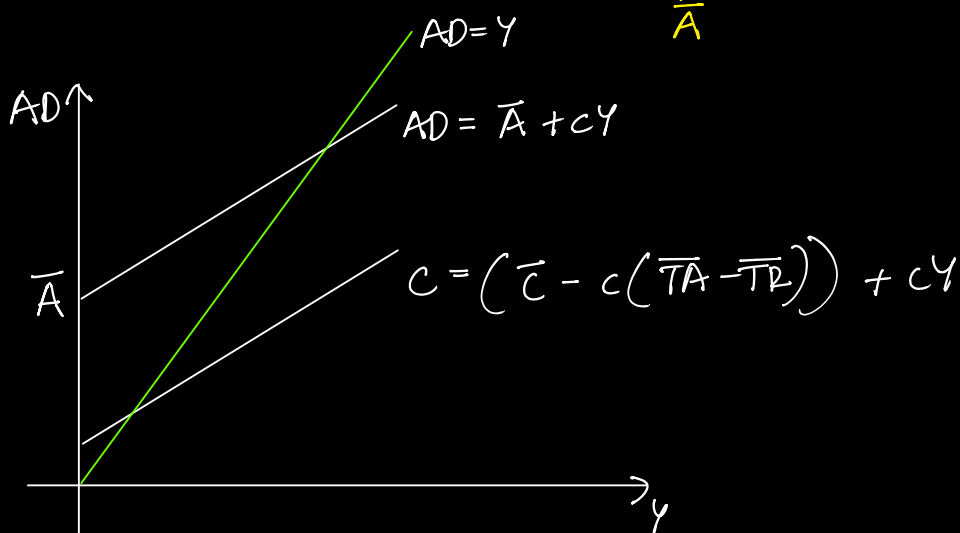
$$\begin{aligned} Y_D &= Y - \bar{T}_A + \bar{T}_R, \quad Y: \text{Income} \\ C &= \bar{C} + cY_D \\ &= \bar{C} + c(Y - \bar{T}_A + \bar{T}_R) \end{aligned}$$

Y_D : Disposable income

$$AD = C + I + G + NX$$

$$= \bar{C} + c(Y - \bar{T}_A + \bar{T}_R) + \bar{I} + \bar{G} + \bar{N}_X$$

$$= (\bar{C} - c\bar{T}_A + c\bar{T}_R + \bar{I} + \bar{G} + \bar{N}_X) + cY$$



Sold (Consumed)

Saving

$$Y = C + I = C + S$$

Output
(Production)

Invested (Stored)

??

$$(S = I)$$

For a simplistic economy

Complex economy with govt, foreign trade, ...

isolated / brain / underinventory

$$Y = C + I + G + NX$$

$$Y_D = Y + TR - TA \Rightarrow Y_D = C + I + G + NX + TR - TA$$

$$Y_D = C + S$$

$$C + S = C + I + G + NX + TR - TA$$

$$S = I + G + NX + TR - TA$$

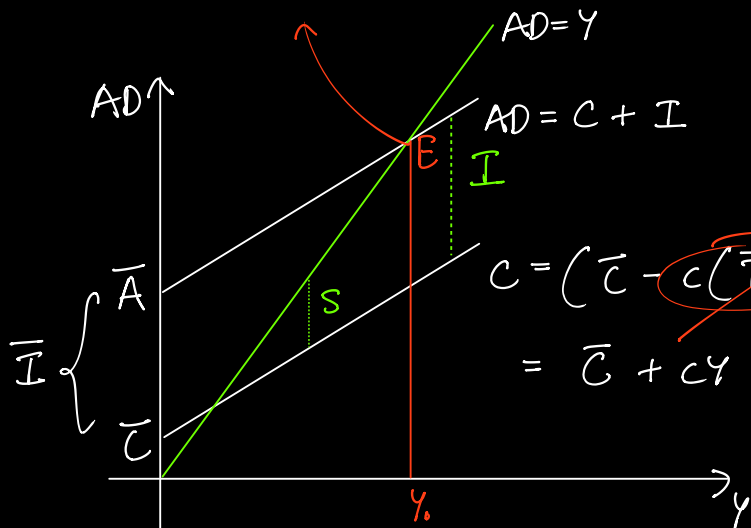
$$= I + (G + TR - TA) + NX$$

Budget Deficit (BD) > 0

Output of the economy as well as well as income of a household, they are used interchangeably

Only at eq. point
 $S = I$

Savings by households are used to clear BD.



0 as there is no government

$$C = (\bar{C} - c(TA - TR)) + cY$$

$$= \bar{C} + cY$$

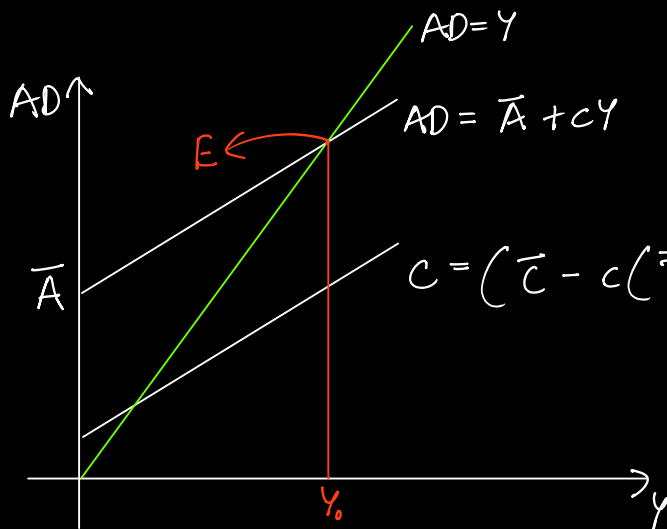
$$Y = C + S + TA - TR \Leftrightarrow \begin{cases} Y_D = Y + TR - TA \\ Y_D = C + S \end{cases}$$

$$Y = C + I + G + NX$$

$$I = S + (TA - TR - G) - NX$$

24/1/25

Consumption and Output



$$AD = C + \bar{I} + \bar{G} + \bar{NX}$$

$$C = (\bar{C} - c(\bar{T}_A - \bar{T}_R)) + cY$$

Autonomous Expenditure

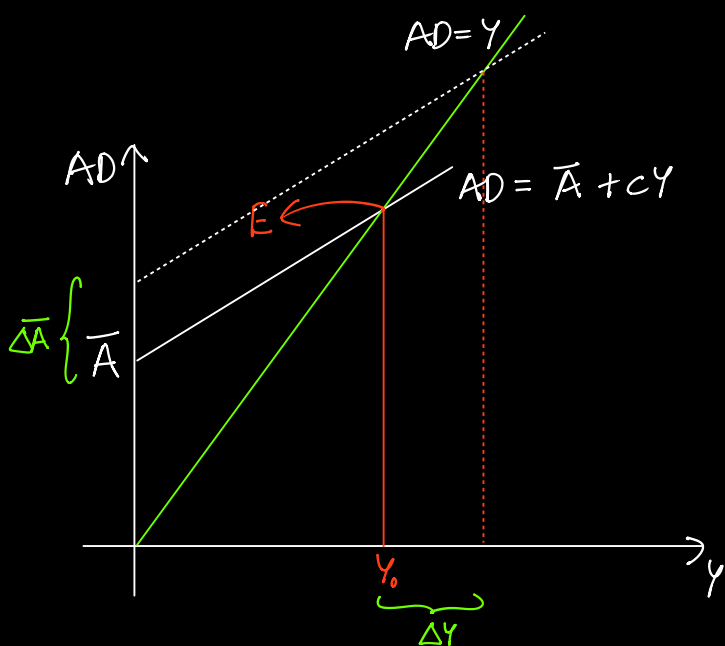
$$Y = AD$$

$$Y = \bar{A} + cY \Rightarrow Y_0(1-c) = \bar{A}$$

$$Y_0 = \frac{\bar{A}}{1-c} \Rightarrow \Delta Y_0 = \frac{\Delta \bar{A}}{1-c}$$

Keynesian Multiplier

Eg:- $c=0.9$
 $\Delta \bar{A}=1$
 $\Delta Y_0=10$
 magnifying effect



$\bar{A}$ Autonomous expenditure
 Demand

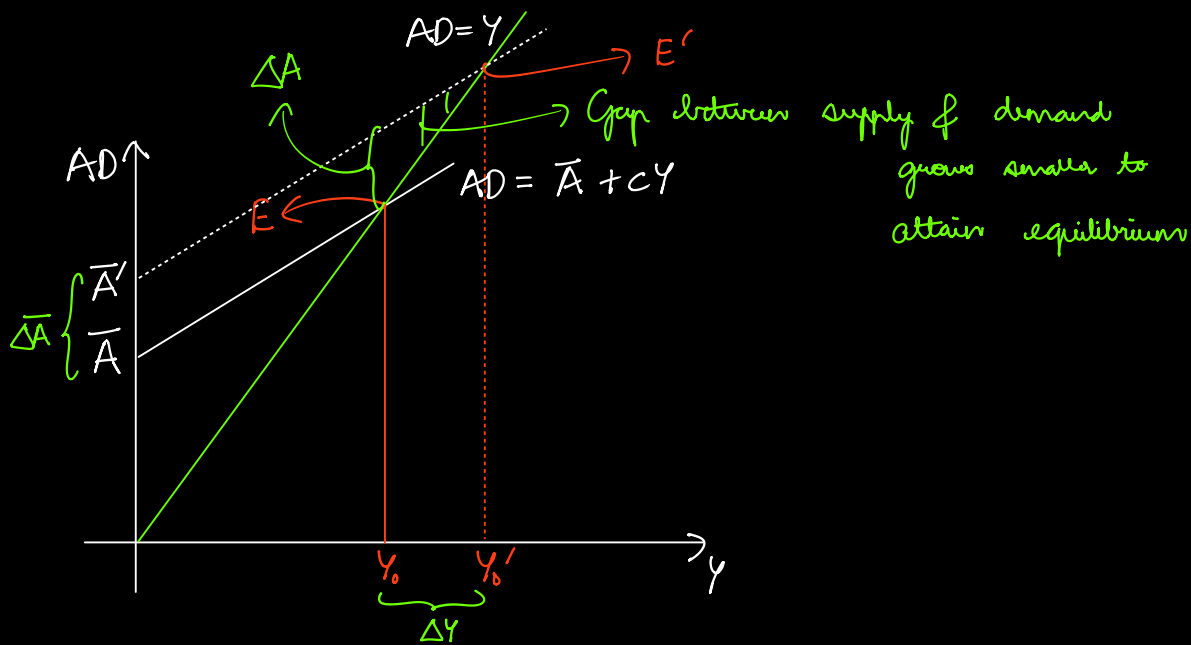
Eg:- $\uparrow$ autonomous spending by $\$1$

By how much does equilibrium increase?

Round	$\uparrow$ in demand	$\uparrow$ in production	$\uparrow$ in income
1	ΔA	ΔA	ΔA
2	$c \Delta A$	$c \Delta A$	$(1+c) \Delta A$
3	$c^2 \Delta A$	$c^2 \Delta A$	$(1+c+c^2) \Delta A$
$\vdots$			$\vdots$

$$\frac{\Delta A}{1-c}$$

$$\begin{aligned} \Delta Y_0 = \Delta AD &= \Delta A (1 + c + c^2 + \dots) \\ &= \frac{\Delta A}{1-c} \end{aligned}$$

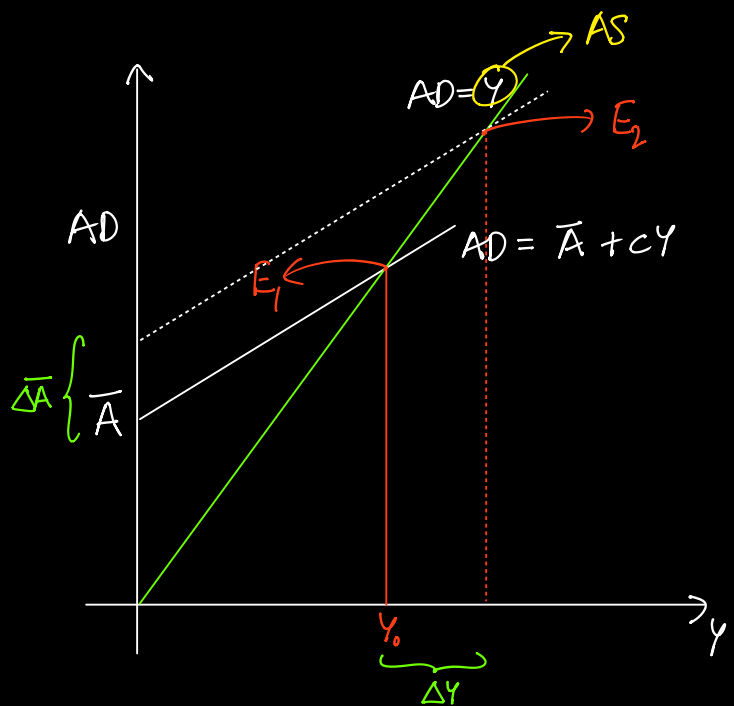


29/1/25

Keynesian Multiplier

$$\Delta Y = \alpha \Delta \bar{A}$$

$$\left(\frac{1}{1-c} \right)$$



Fiscal Policy
G, TA, TR

$$Y_D = Y + TR - TA, \quad C = \bar{C} + cY_D$$

$$C = \bar{C} + c(Y + TR - TA)$$

$$= \bar{C} + c(Y + TR - tY)$$

Disposable income

Taxes
dependent on income

Taxes have an effect on
MPC

$$= \bar{C} + cTR + c(1-t)Y$$

Marginal Propensity to Consume has gone down

$$AD = C + I + G + NX$$

$$= (\bar{C} + c\bar{T}R + c(1-t)Y) + \bar{I} + \bar{G} + \bar{N}X$$

$$= (\bar{C} + c\bar{T}R + \bar{I} + \bar{G} + \bar{N}X) + c(1-t)Y$$

$$= \bar{A} + \underbrace{c(1-t)Y}$$

slope of graph goes down

$$AD = \bar{A} + c(1-t)Y$$

$$Y = AD$$

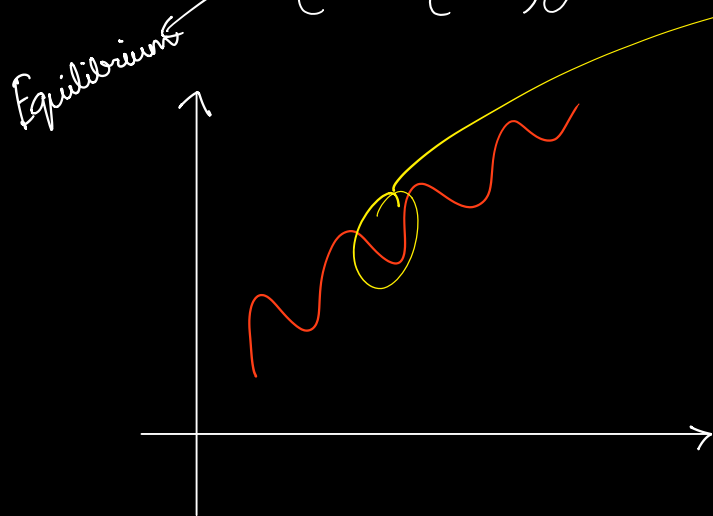
$$Y = \bar{A} + c(1-t)Y$$

$$Y^*(1 - c(1-t)) = \bar{A}$$

Expansionary policies increase $\bar{A}$

Taxes helps prevent ups & downs
(fluctuations)

Automatic stabilisers



30/1/25

Automatic Stabilisers

→ Income Tax
→ Subsidies

$G, \bar{T}A, \bar{T}R$

affect

Budget surplus

Government
Revenue

↓ ↑ ↓

$$\Delta Y = \alpha_G \Delta \bar{G}$$

$$\alpha_G = \frac{1}{1 - c(1-t)}$$

$$\bar{G}_1 \uparrow, \bar{T}R \uparrow \Rightarrow c \Delta TR$$

$$\Rightarrow \alpha_{G_1} c \Delta TR$$

t : Tax rate $\uparrow$

$$Y_D \downarrow \Rightarrow C \downarrow$$

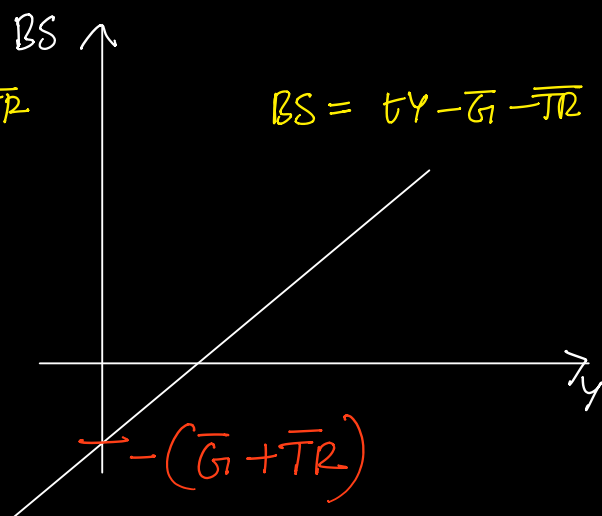
$$\Rightarrow AD \downarrow$$

$$\text{Budget Surplus (BS)} = TA - \bar{G}_1 - \bar{T}R$$

$$BS = tY - \bar{G}_1 - \bar{T}R$$

$$BS = tY - \bar{G}_1 - \bar{T}R$$

Per unit $\leftarrow$ taxation



$$G_1 \uparrow \Rightarrow \Delta Y^* = \alpha_{G_1} \Delta \bar{G}_1$$

$$\Delta TA = t \alpha_{G_1} \Delta \bar{G}_1$$

$$\Delta BS = \Delta TA - \Delta \bar{G}_1$$

$$= t \alpha_{G_1} \Delta \bar{G}_1 - \Delta \bar{G}_1 = \frac{(1-c)(1-t)}{1-c(1-t)} \Delta \bar{G}_1$$

Balance Budget Multiplier

$$\Delta G_1 \uparrow, TA \uparrow$$

$$\Delta G_1 = \Delta TA$$

$$\Delta BS = 0$$

$$\Delta Y > 0$$

$$G_1 \uparrow \Rightarrow \Delta Y = \alpha \Delta G_1$$

$$TA \uparrow \Rightarrow Y_D \downarrow \Rightarrow C \downarrow$$

$$C \downarrow \text{ by } c \Delta TA \Rightarrow Y \downarrow$$

$$Y \downarrow \text{ by } \alpha c \Delta TA$$

$$\begin{aligned}\Delta Y &= \alpha \Delta G - \alpha c \Delta G \\ &= \cancel{\alpha} \Delta G (1 - c) \quad \left(\alpha = \frac{1}{1-c} \right) \\ &= \Delta G\end{aligned}$$

For balance $\Delta Y = \Delta G = \Delta TA$

Q. $c = 0.8$

$\Delta G = 200$, $\Delta TA = 200$

$\Delta Y^* = ?$

ΔAD due to ΔG



$$\begin{aligned}\Delta AD &= \alpha \Delta G; \alpha = \frac{1}{1-c} \\ &= \frac{1}{1-0.8} \times 200\end{aligned}$$

$= 1000$

↗ consumption

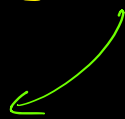
↓ in expenditure $= 0.8 \times 200$
 $= 160$

↓ in eq output $= 160 \times 5 = 800$

$\Delta TA = \Delta G$

$$\Delta Y = \frac{1}{1-c} \left(\Delta G (1-c) \right) = \Delta G = \Delta TA = 200$$

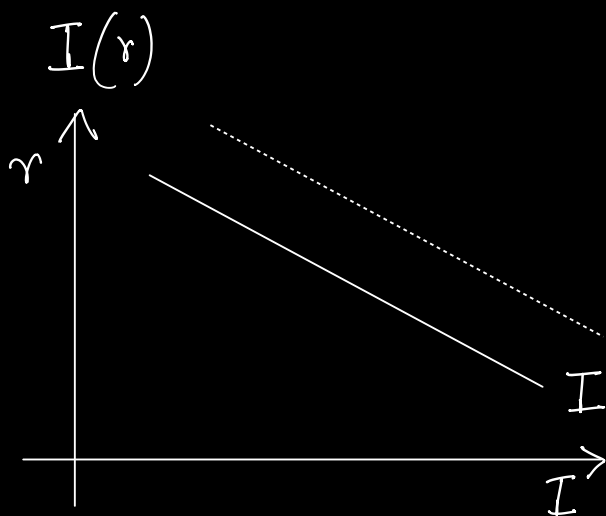
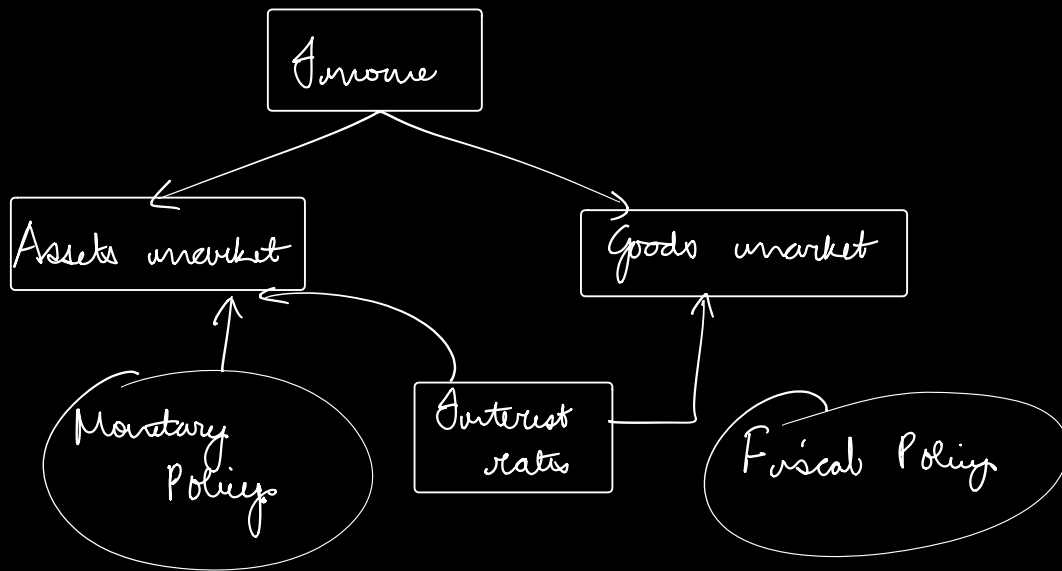
$$\frac{1}{1-c} (\Delta G - c \Delta TA)$$



$$\begin{aligned}AD &= C + I + NX + G \\ &= C + I(r) + G + NX\end{aligned}$$

$I(r)$: I as a func of r (interest rate)

$(Y^*, r^*) \rightarrow$ Joint eq. (both goods market and money market get cleared)



$$I = \bar{I} - br$$

$$b > 0$$

$$AD = C + I + G + NX$$

$$C = \bar{C} + cY_D$$

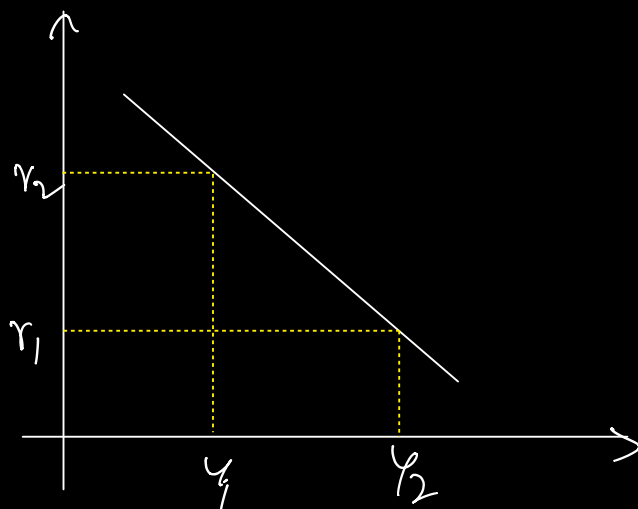
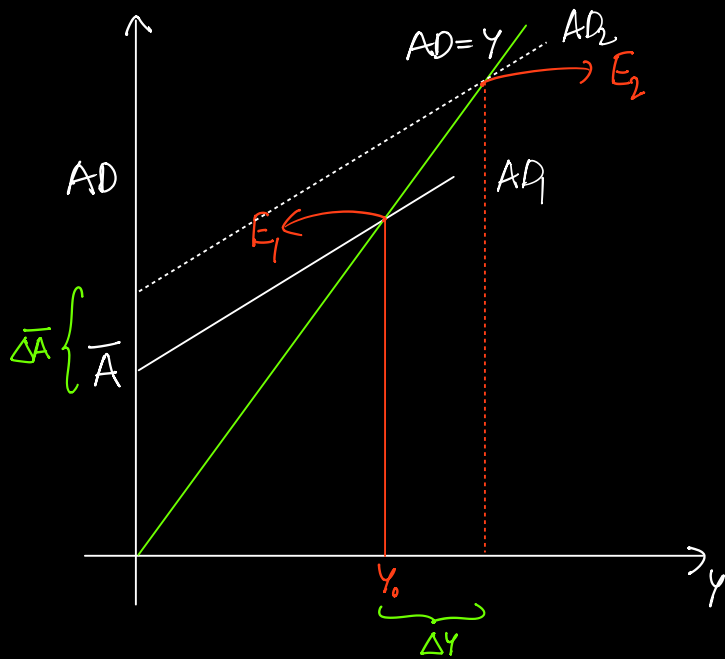
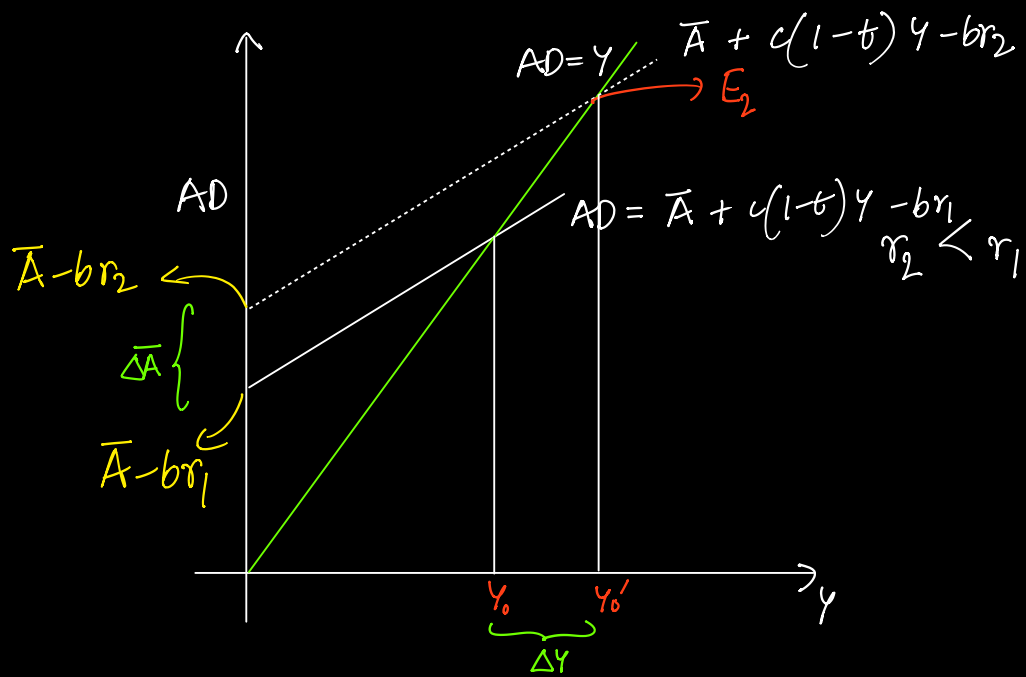
$$= \bar{C} + c(Y + TR - tY)$$

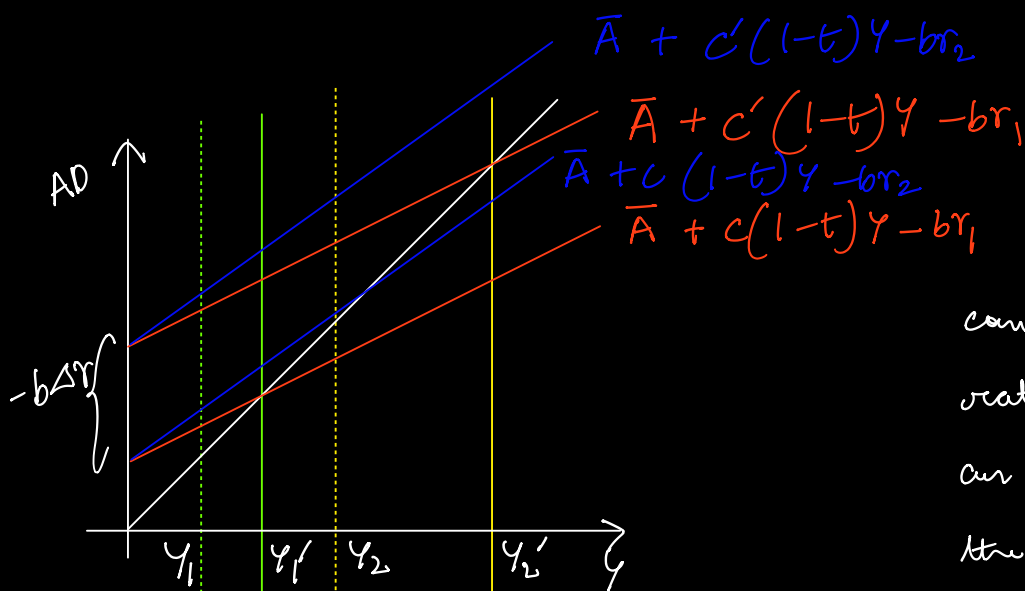
$$= \bar{C} + c\bar{TR} + c(1-t)Y$$

$$AD = \bar{C} + c\bar{TR} + c(1-t)Y + (\bar{I} - br) + \bar{G} + \bar{NX}$$

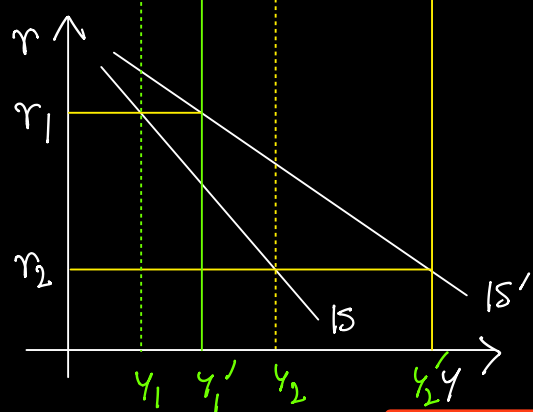
$$AD = \bar{A} + c(1-t)Y - br$$

$$\bar{A} = \bar{C} + c\bar{TR} + \bar{I} + \bar{G} + \bar{NX}$$





IS curve - plotting all combinations of interest rate and output for an equilibrium condition in the goods market.



$$Y = AD = \bar{A} - br + c(1-t)Y$$

$$= \alpha_G (\bar{A} - br)$$

$$\alpha_G = \frac{1}{1-c(1-t)}$$

IS schedule

$$Y = \alpha_G (\bar{A} - br)$$

Interest rate
↑ Tradeoff
Demand/Output

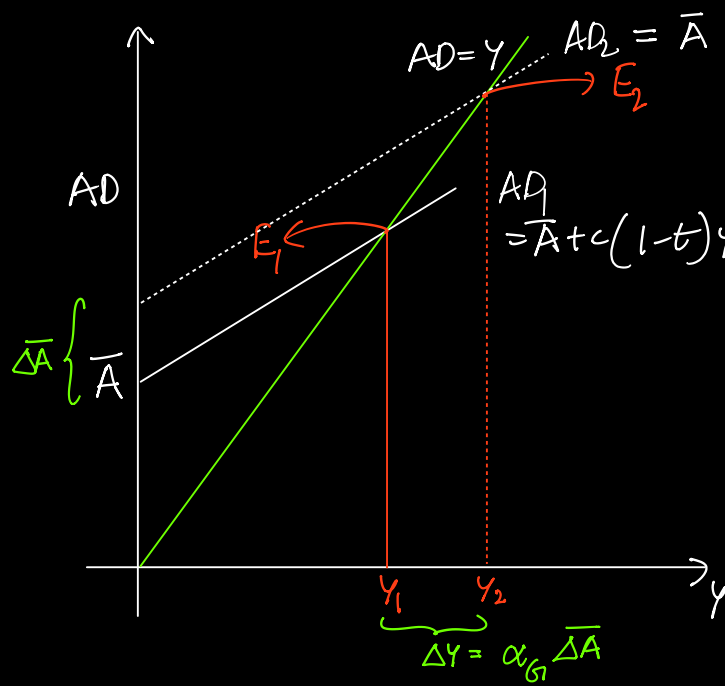
SLOPE OF IS CURVE

$$\frac{\partial r}{\partial Y} = \frac{-1}{\alpha_G b}$$

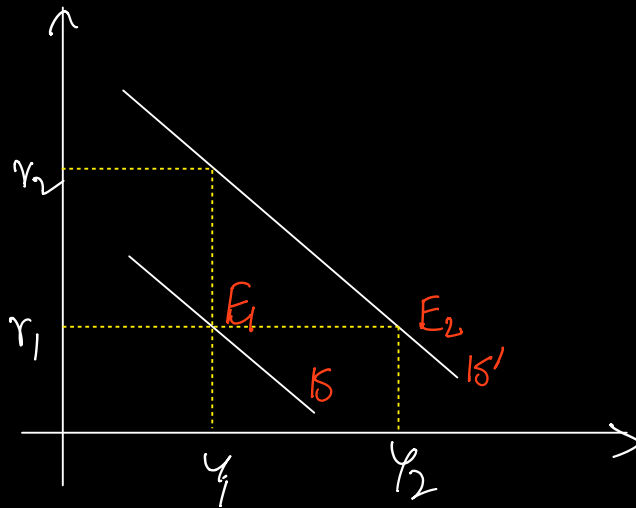
Large $\rightarrow I = \bar{I} - br$
(slope)
Flatter IS schedule

Fiscal Policy $\rightarrow t$ (tax rate)

The entire discussion above was in the context of the goods market.



intercept
 ↑
 Change in position - factors
 other than interest changes
 Change in slope
 ↓ interest rate



7/2/25 (i and r are both interest rates)

MONEY MARKET EQUILIBRIUM

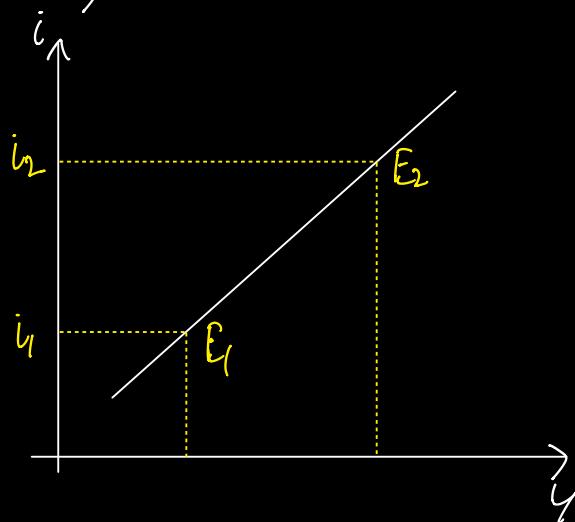
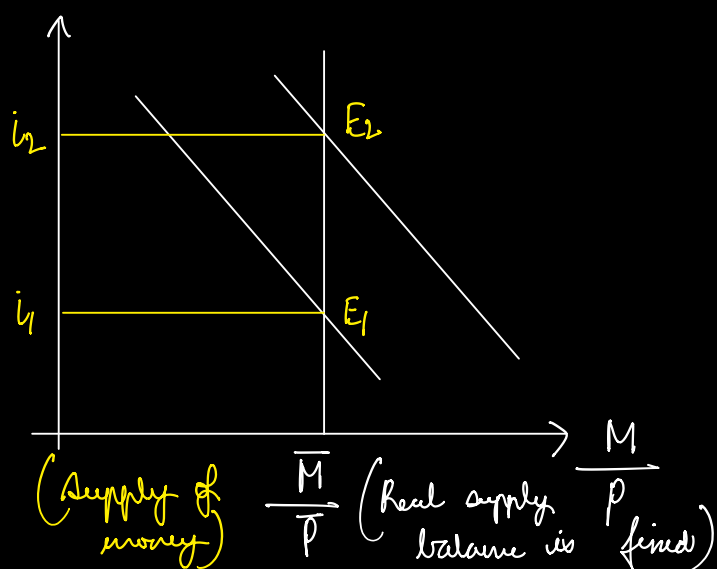
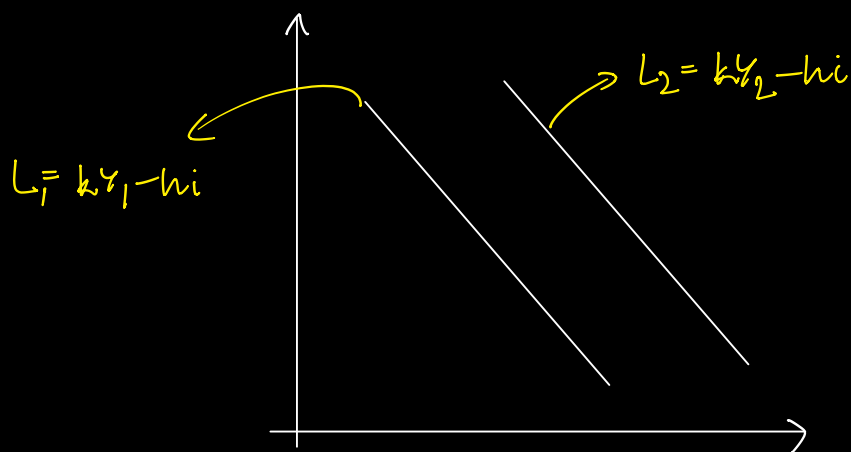
(i, Y) $\left\{ \begin{array}{l} \text{Demand for money} \\ \text{Real (Demand for real balances)} \end{array} \right.$ $\rightarrow$ Amount of goods & services that can be purchased with the money.

	A	B
Nominal money	100	200
Nominal price	2	4
Real balance (Purchasing power)	50	50

$$L = kY - hi$$

Real balance (demand for money)

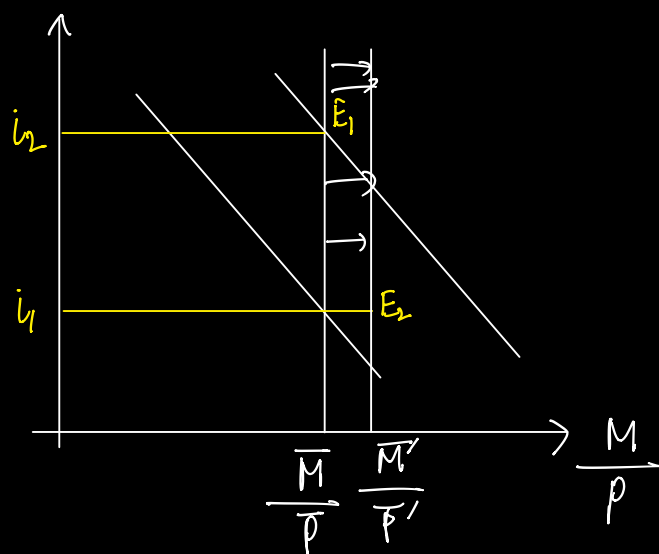
Demand for money - Demand for the amount of real goods and services money can buy.



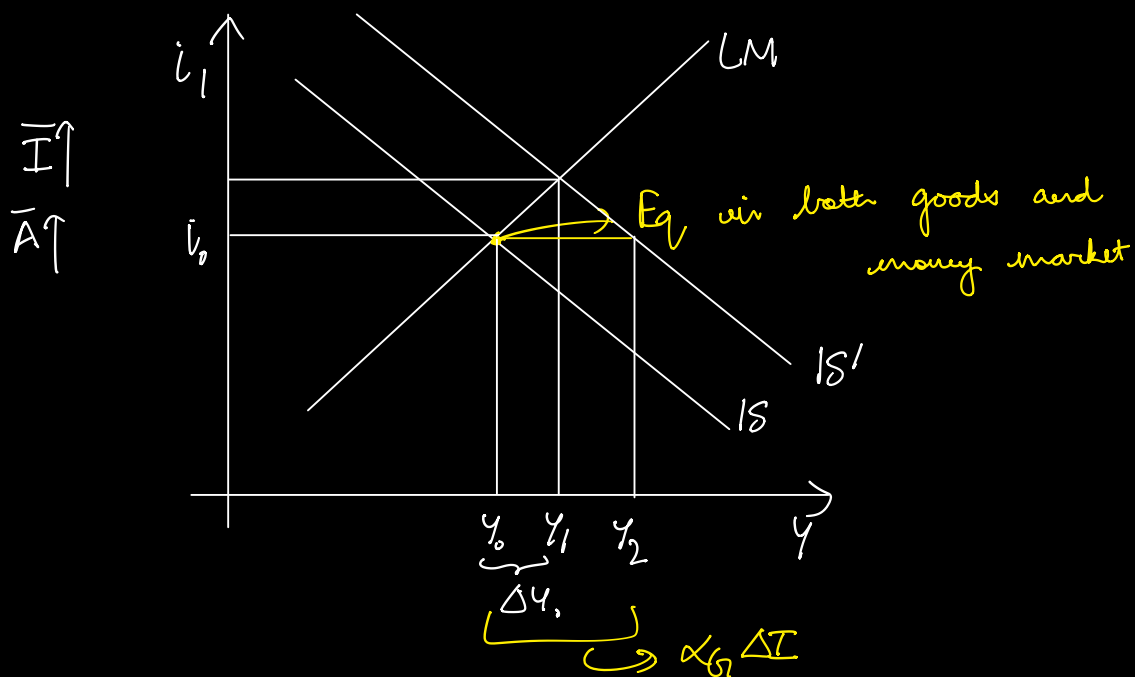
$$\frac{\bar{M}}{P} = kY - hi$$

$$i = \left(\frac{\bar{M}}{P} - kY \right) / h$$

LM curve $\Rightarrow \frac{\partial i}{\partial Y} = -\frac{k}{h}$



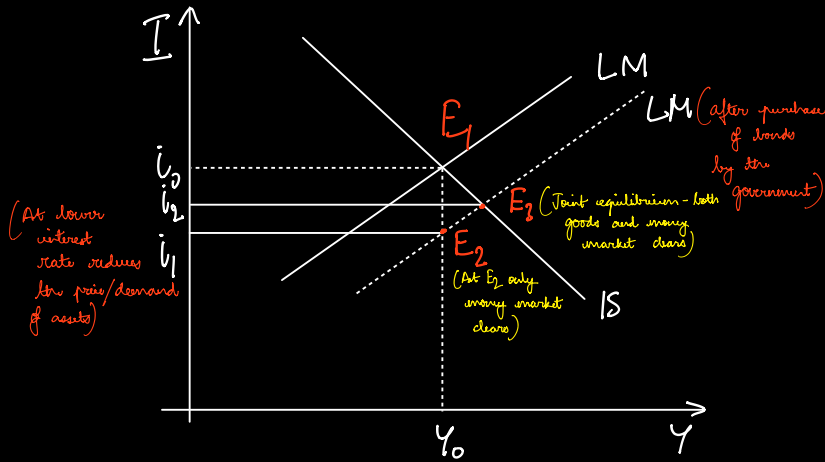
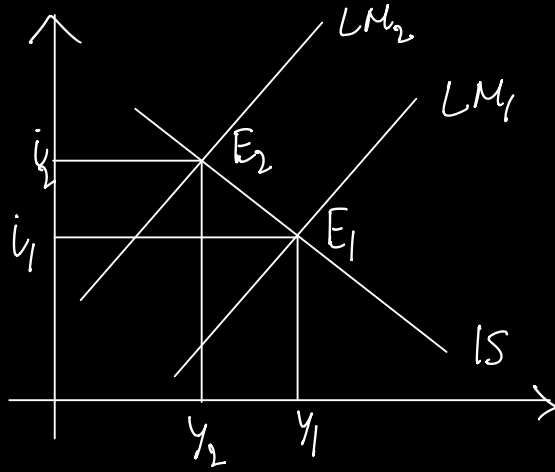
Joint Eq in both Money and Goods Market



$\Delta Y_0 < \alpha_G \Delta I$ as interest rate has also increased

$$P_1 \rightarrow P_2$$

$$\left(\frac{M}{P}\right) \downarrow$$



Assets market - monetary policy
 Goods market - Fiscal policy

open market operations

Transaction of bonds

contract

Eg:- Holder receives a certain amount of money after a certain period of time.

Economy moves from $E_1 \rightarrow E_2 \rightarrow E_3$

Purchasing bonds injects money into the market

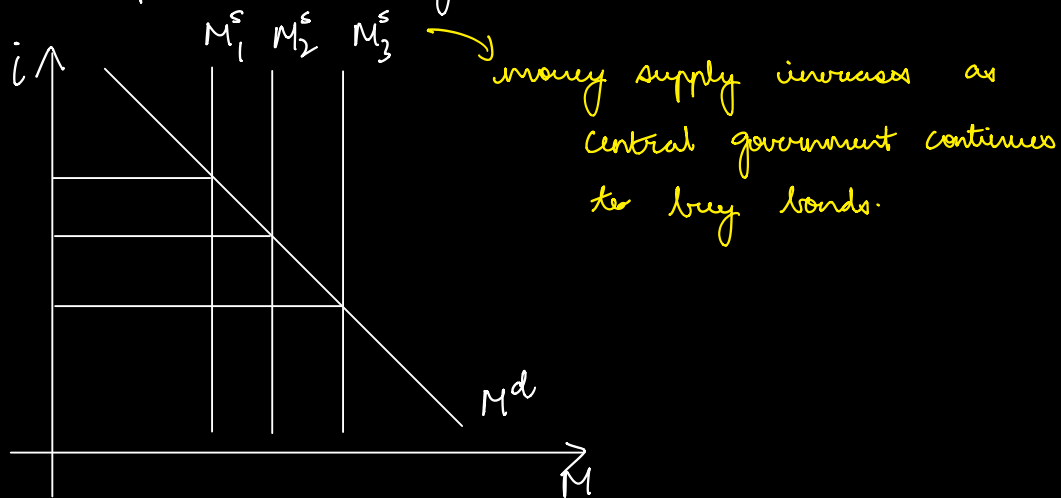
money supply increases

→ reduction of interest rate

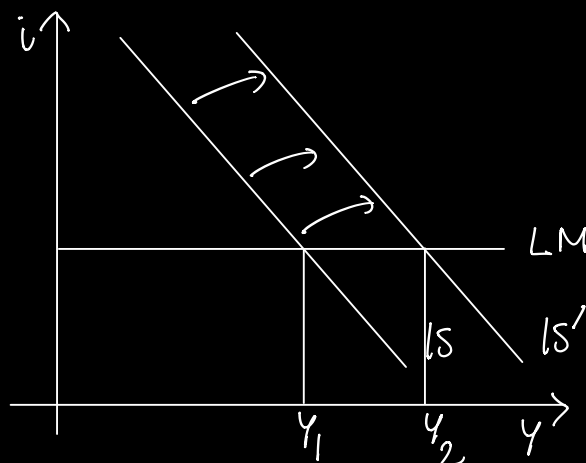
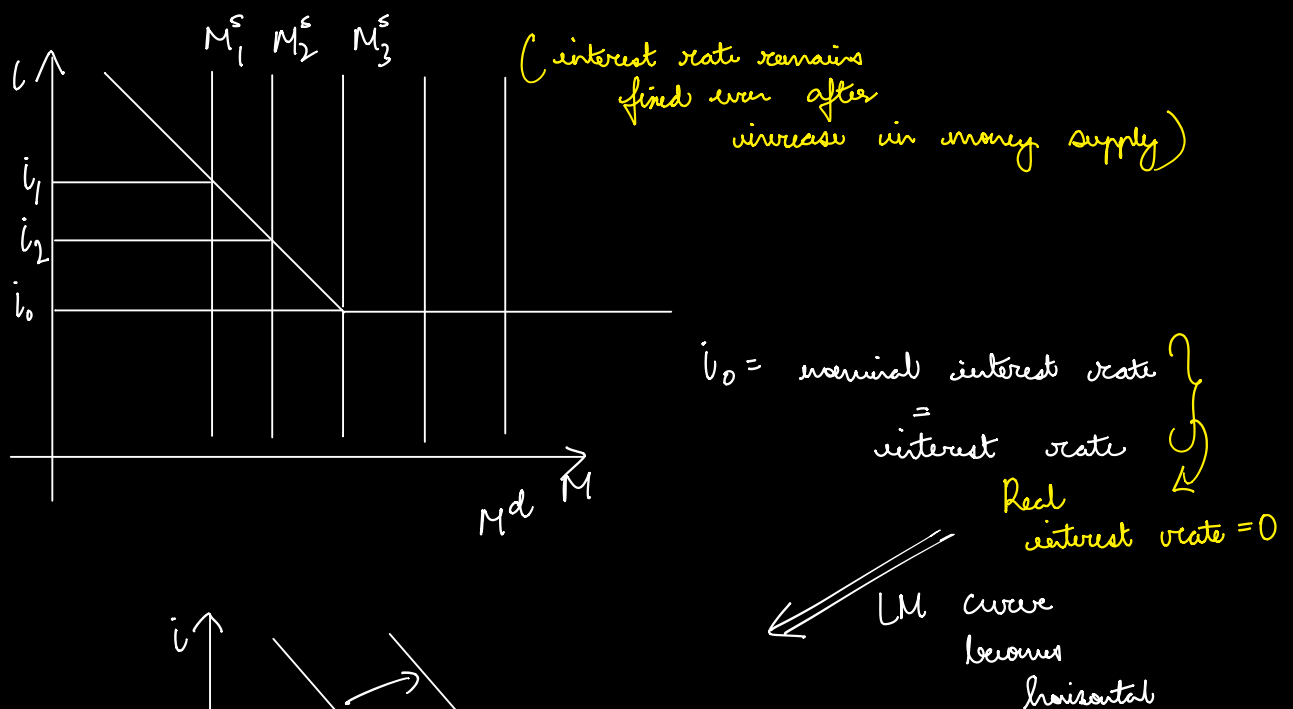
$$M^d = f(i, Y); \quad i \downarrow, Y \uparrow$$

LIQUIDITY TRAP

During a recession cutting down of interest rates does not necessarily work to uplift the economy.



Real interest rate = nominal interest rate - inflation



No crowding as interest rate is constant. Investment does not fall.

Continuation: Government spending more money

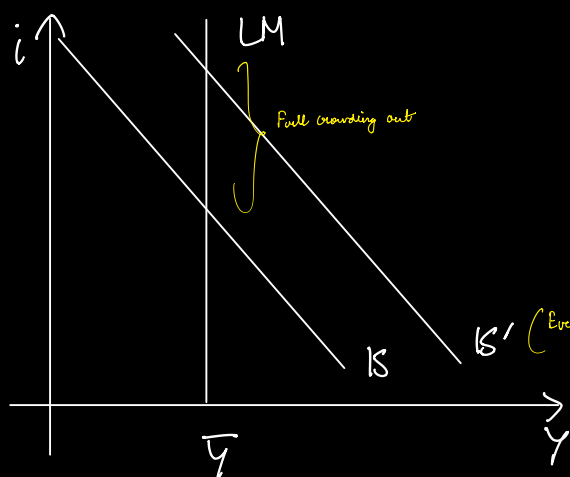
↓
more jobs

↓
ability to spend ↑

↓
demand ↑

↓
inflation ↑ (caused by government's
expansive fiscal policy)

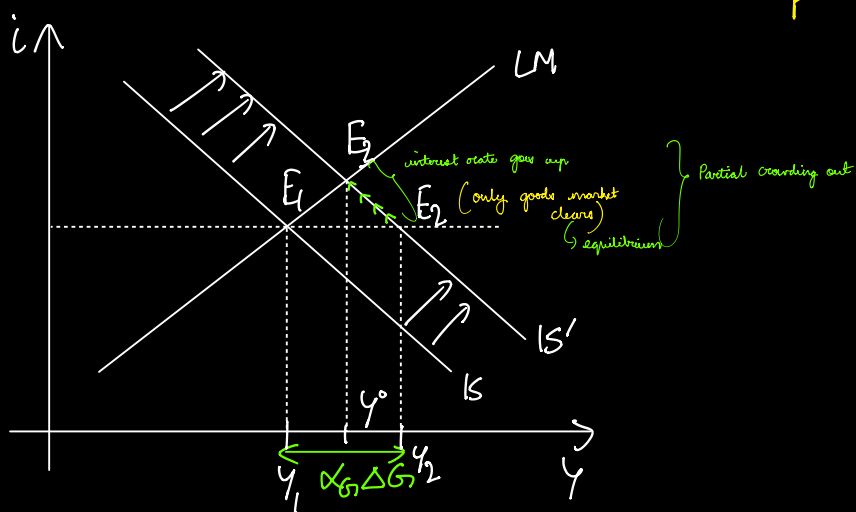
same inflation
is fine for
economy to get back

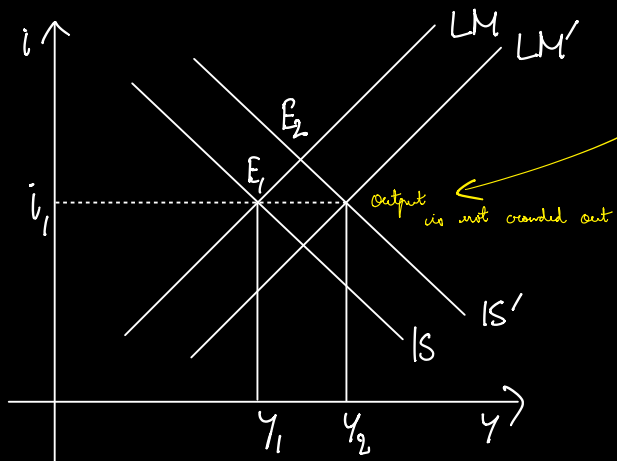
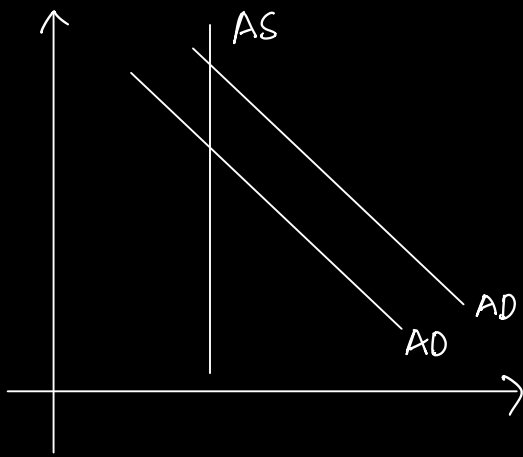


$$\left(\frac{M}{P}\right)^s = kY - hi; \quad h=0$$

(money demand
is not sensitive to
interest rate)

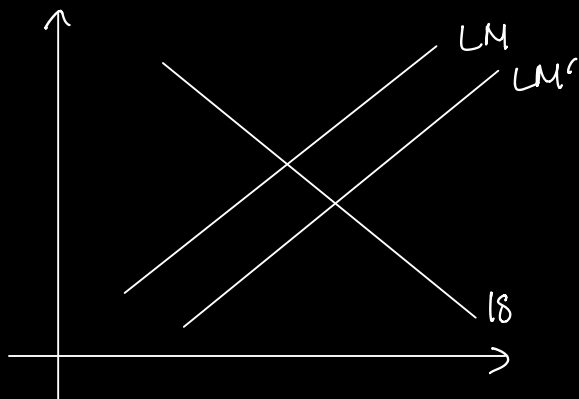
$$\frac{\bar{M}}{\bar{P}} = kY \Rightarrow \boxed{M = PkY}$$



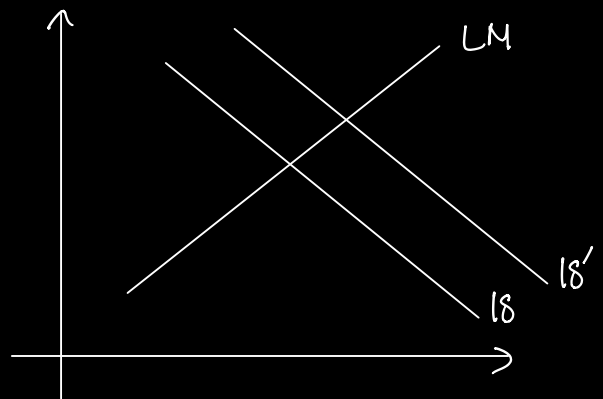


Fully accommodative monetary policy $\Rightarrow$ Monetising budget deficit

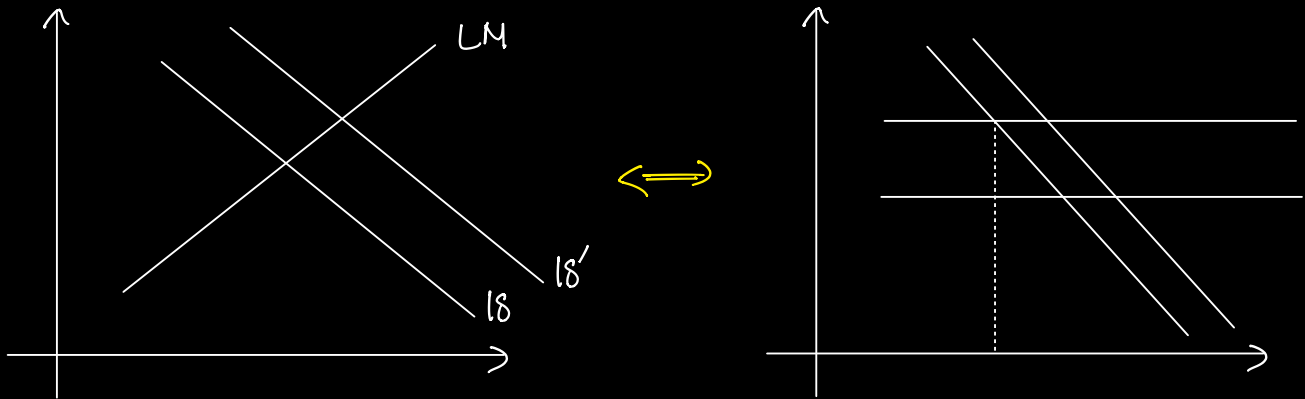
Monetary Policy $\uparrow$



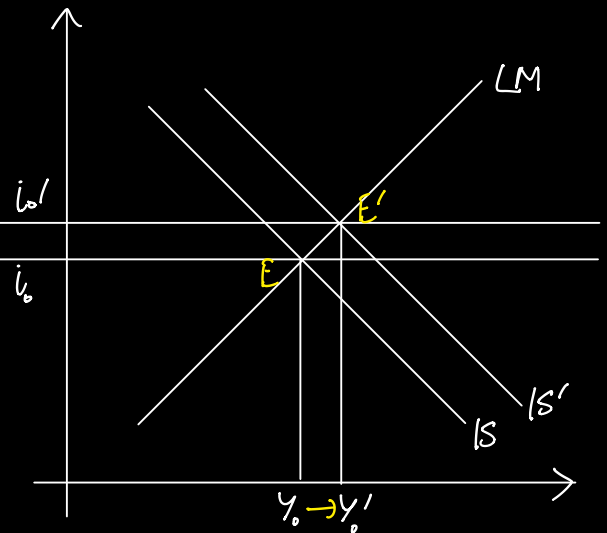
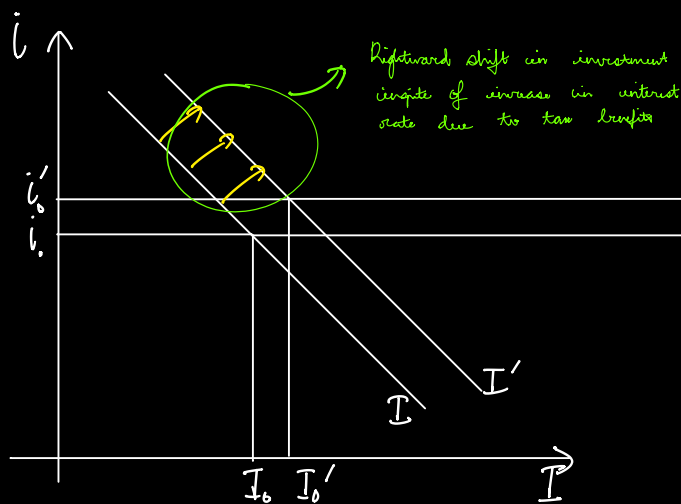
Fiscal Policy $\uparrow$



Investment tax credits subsidise investment, effect on investment by increase in tax is negated.

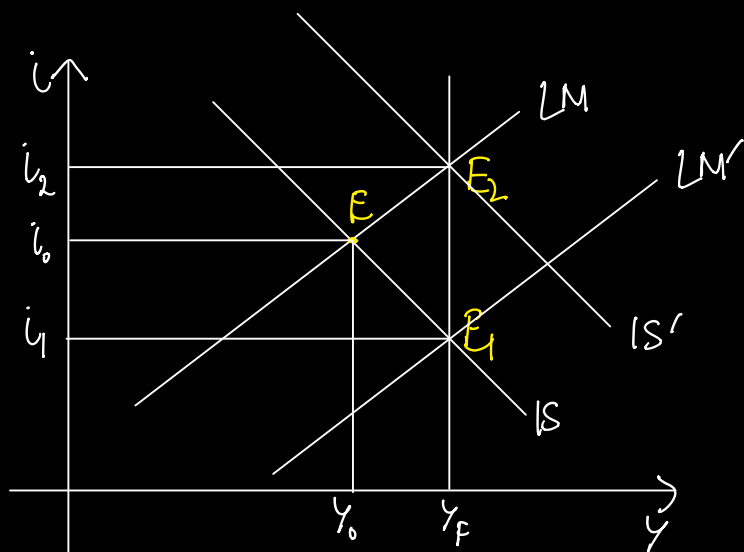


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Alternative Fiscal Policies

	i	C	I	Y
Income Tax Cut	+	+	-	+
Govt Spending	+	+	-	+
Investment Subsidy	+	+	+ / or stay the same	+



Economy is never at full employment level, there are unutilised resources.

Y_F : Full employment

E_1, E_2 : Two equilibrium paths to attain full employment.

$$IS: Y = \alpha_G (\bar{A} - bi) \quad ; \quad \alpha_G = \frac{1}{1 - c(1-t)}$$

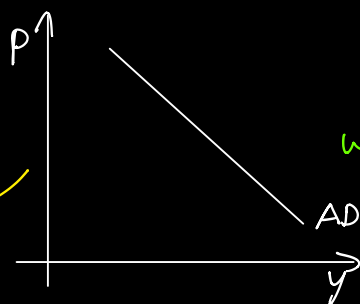
$$LM: i = \frac{1}{h} \left(kY - \frac{\bar{M}}{\bar{P}} \right)$$

Equilibrium in the goods market

$$Y = \sigma \bar{A} + \sigma \frac{b}{h} \frac{\bar{M}}{\bar{P}} \quad ;$$

$$\sigma = \frac{\alpha_G}{1 + k\alpha_G b/h}$$

Increase in any component of $\bar{A}$ from with constant i can lead to same effect



$$\sigma = \frac{\alpha_G}{1 + k\alpha_G b/h}$$

$$\frac{dY}{dG} = \sigma = \frac{h\alpha_G}{h + kb\alpha_G}$$

$$i = \frac{k}{h} \sigma \bar{A} - \frac{1}{h\alpha_G} \sigma \frac{\bar{M}}{\bar{P}}$$

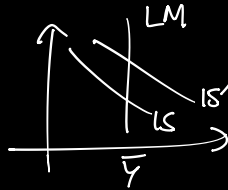
$$\gamma < \alpha_G$$

 $\Rightarrow$

$$\frac{1}{1 + \frac{h\alpha_G b}{h}} < 1$$

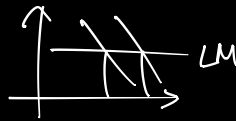
I

$$h \rightarrow 0, \gamma \rightarrow 0$$



II

$$h \rightarrow \infty, \gamma \rightarrow \alpha_G$$



$$\frac{\partial \gamma}{\partial \left(\frac{M}{P} \right)} = \frac{b}{h} \gamma = \frac{b\alpha_G}{h + kb\alpha_G} \quad \left. \vphantom{\frac{\partial \gamma}{\partial \left(\frac{M}{P} \right)}} \right\} \text{Similar process on money market}$$